Do Now · retrieval

Lesson 155 · PageRank: Iterating a Link Graph

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What does the inverted index map?

Q2

What is the role of a router?

Q3

What does a cache hit mean?

Q4

When can binary search be used?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Build graph — Web pages are vertices; directed hyperlinks are edges.

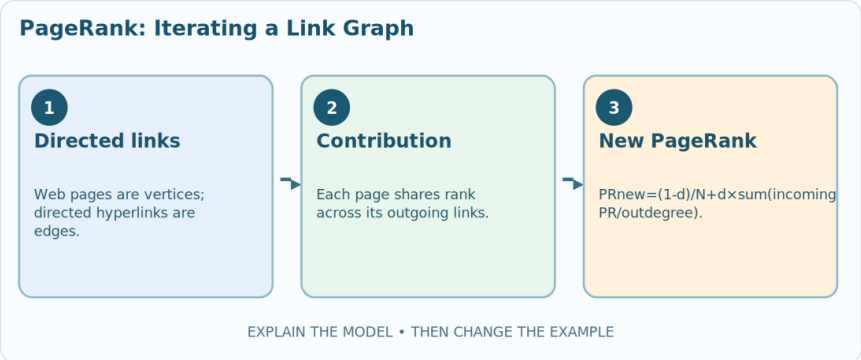
Core idea 2

Distribute rank — Each page shares rank across its outgoing links.

Core idea 3

Apply damping — PRnew=(1-d)/N+d×sum(incoming PR/outdegree).

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

PageRank — Link-based iterative importance algorithm.

directed graph — Nodes joined by one-way edges.

outdegree — Number of a page's outgoing hyperlinks.

damping — Factor balancing link following and random jumping.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

A→B,C; B→C; C→A. Initially PR(A)=PR(B)=PR(C)=1/3; d=0.85. Calculate one iteration.

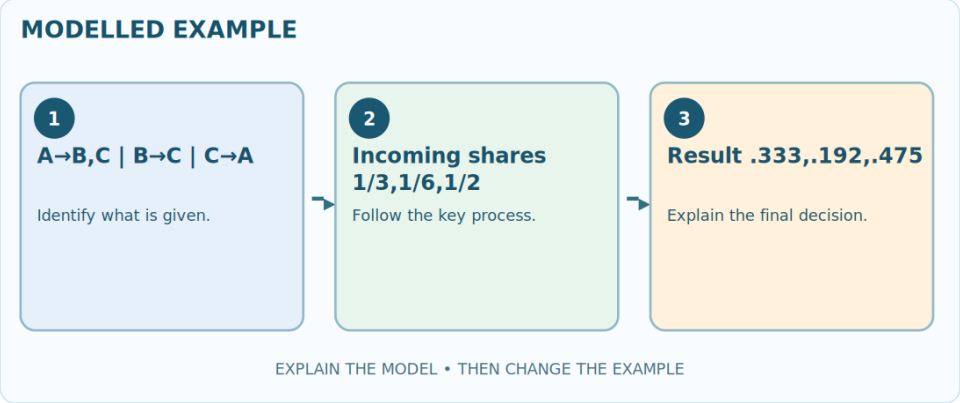
Follow the worked method

1. A→B,C | B→C | C→A — Identify the input or diagram and predict the first result.

2. Incoming shares 1/3,1/6,1/2 — Follow the second stage and explain its importance.

3. Result .333,.192,.475 — Incoming shares A:1/3; B:1/6; C:1/6+1/3=1/2. Base (1-.85)/3=.05. New A=.3333, B=.1917, C=.4750, sum=1.0000.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

A→B | B→A,C | C→A
Identify the input or diagram and predict the first result.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Explain how PageRank uses inbound links and outbound link counts. [6]

Question 2

A page has rank 0.40 and two outgoing links. Calculate each undamped share, explain the division, and recalculate for four outgoing links. [4]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Compute one PageRank update for A→B,C B→C C→A with equal initial ranks and d=.85, then explain an important limitation. [8]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

With PageRank a page with two outgoing links shares its current rank equally between them in the simplified model.

Use key vocabulary from memory

1. Explain PageRank in your own words.

2. Explain directed graph in your own words.

3. Explain outdegree in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.